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CVPR 2023, accepted paper
Kakao Brain Research

Text-grounded Contrastive Learning: Learning to Generate Text-grounded Mask for Open-world Semantic Segmentation from Only Image-Text Pairs

July 16th, 2023

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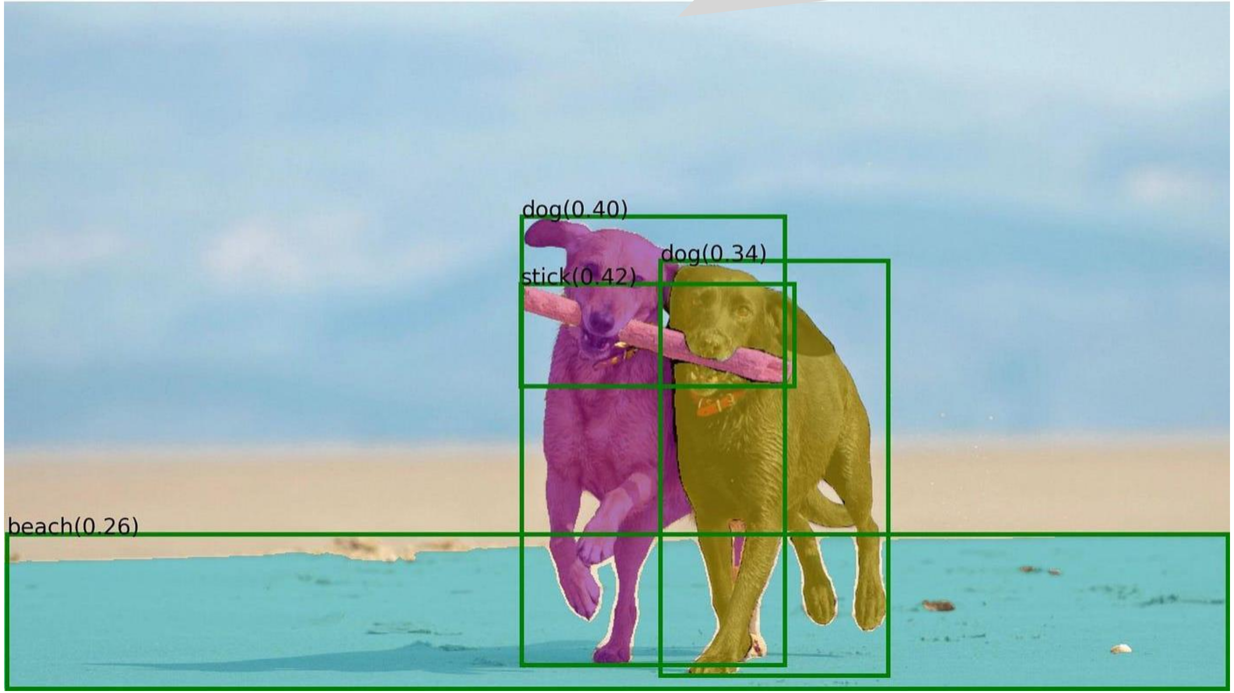
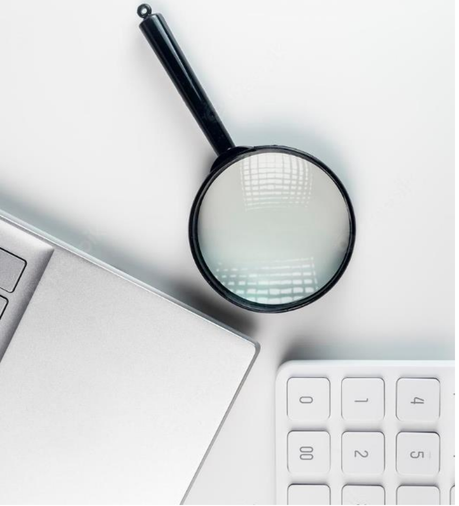
Introduction.

Text-Grounded

There are **two dogs** playing with **a stick** on the **beach**



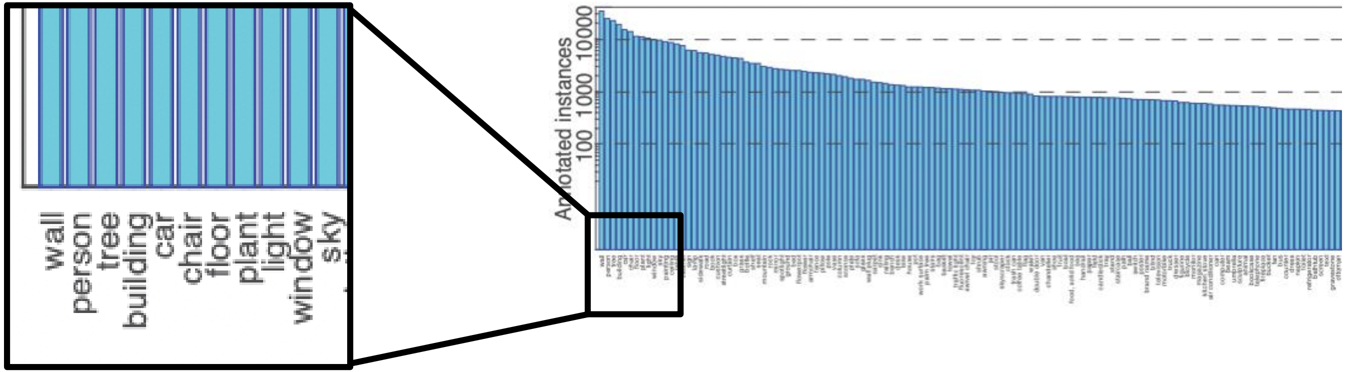
Liu, S. et al (2023). Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. ArXiv. /abs/2303.05499



Open-World (=open-vocabulary)

Closed world (ex. ADE20K)

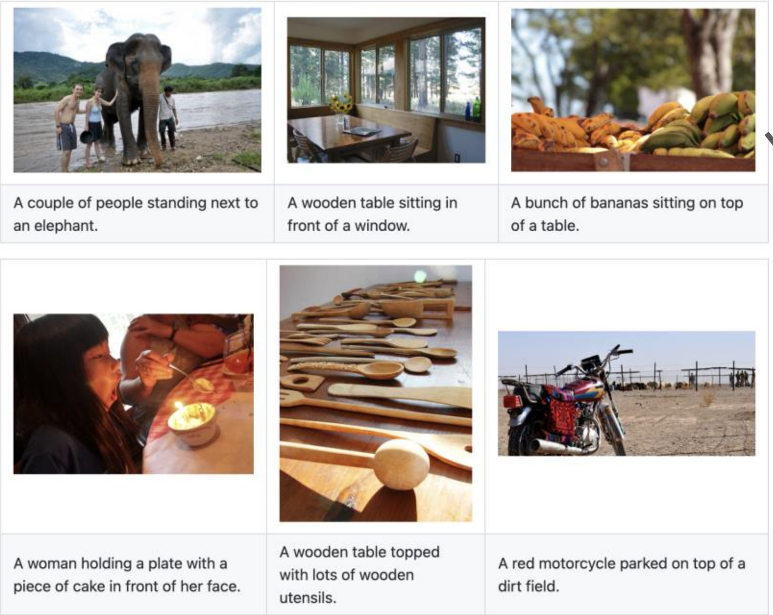
정해져있는 class 를 커버하는 Task



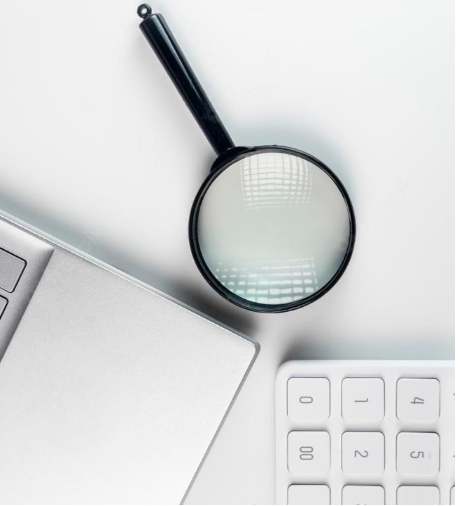
Open-World (=open-vocabulary)

Open World (ex. Web Scratched Datasets)

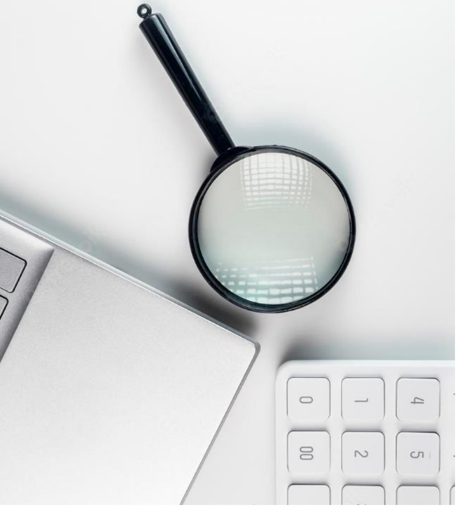
class의 개수가 정해지지 않아 모든 class를 커버하는 Task



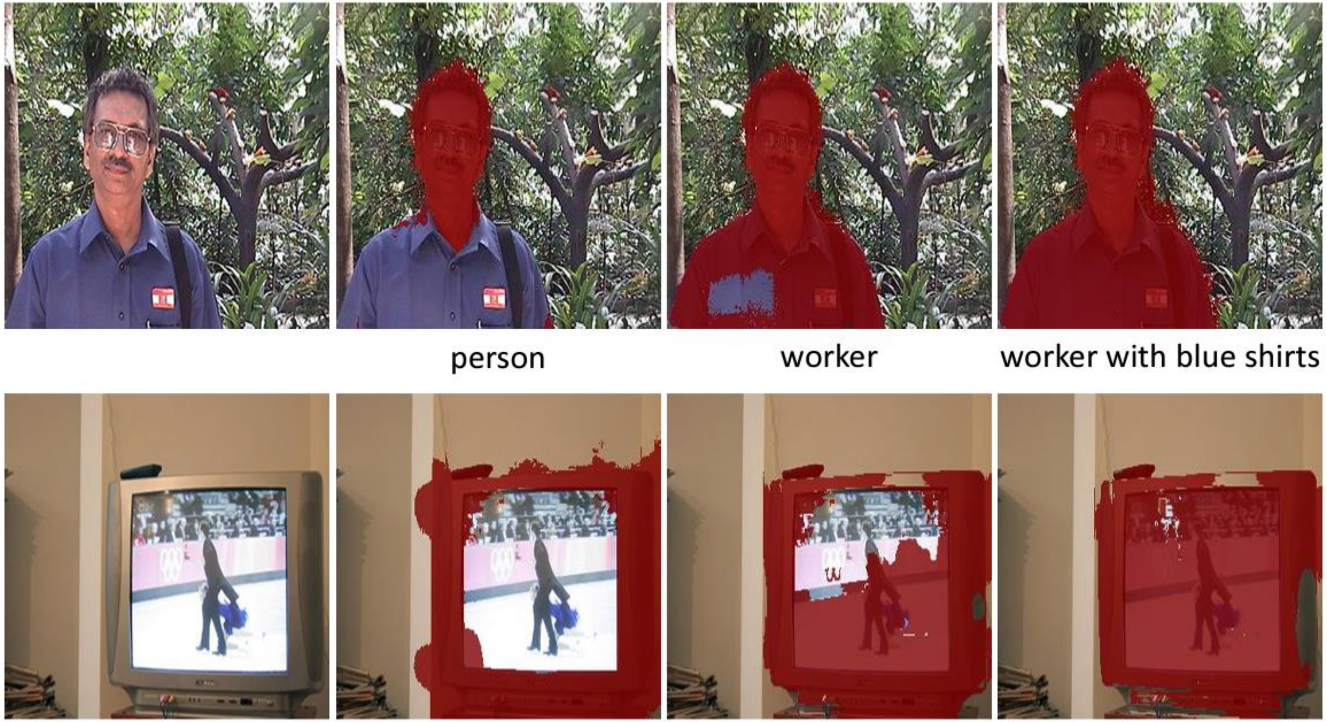
CLIP의 400 million (image, text) pairs 데이터셋



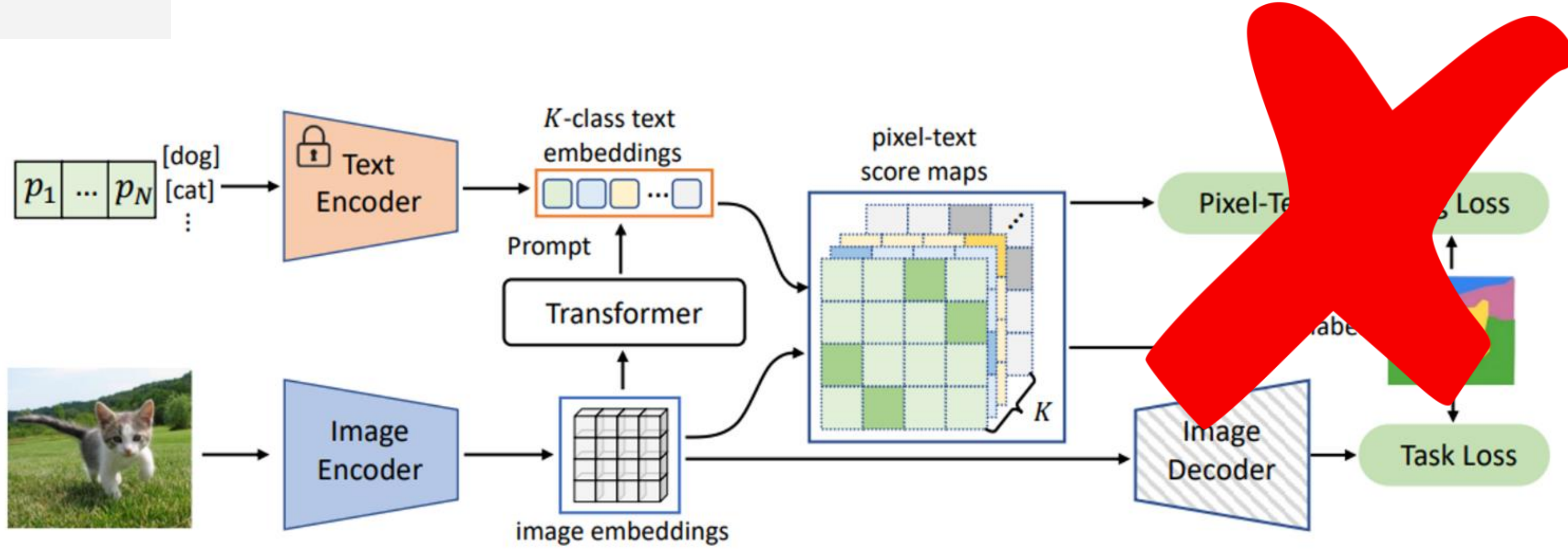
Open-World (=open-vocabulary)



Open-World (=open-vocabulary)



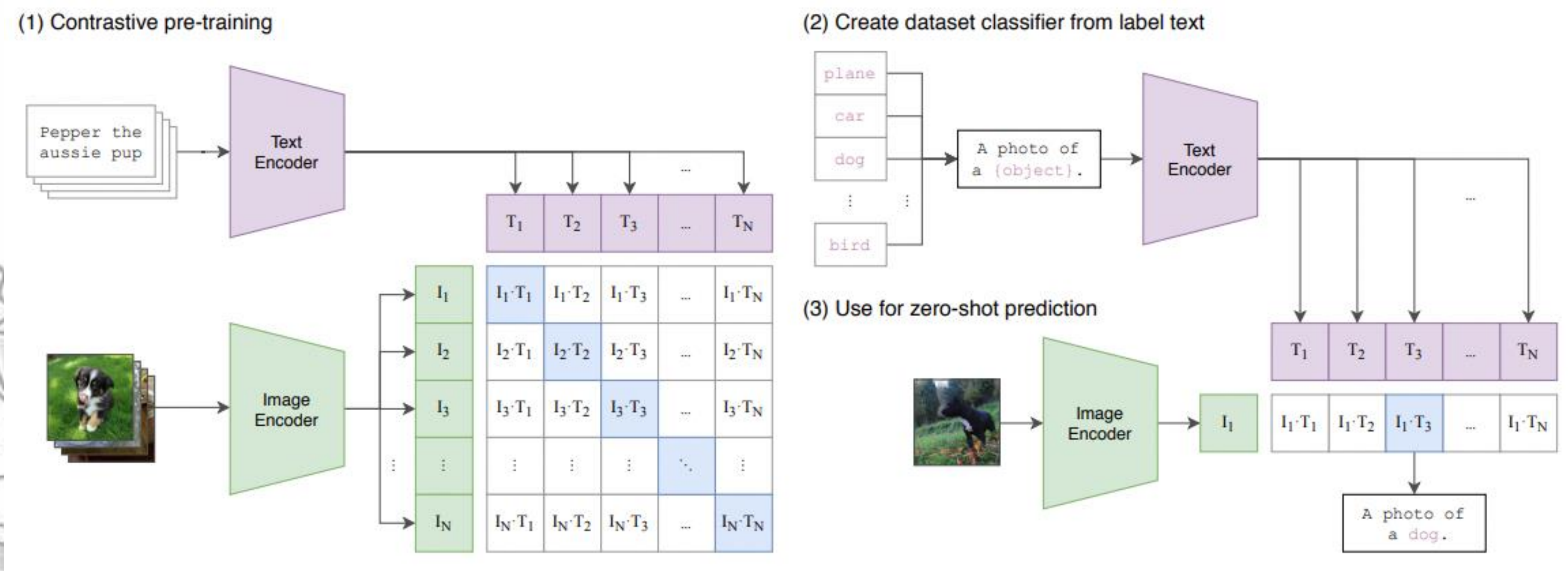
Unsupervised Text-grounded Mask



Rao, Y. et al. (2021). DenseCLIP: Language-Guided Dense Prediction with Context-Aware Prompting. ArXiv. /abs/2112.01518

Related Works

Text-Image embedding alignment: CLIP



Related Works

Semi-supervised Learning

- ✓ Lseg
- ✓ OpenSeg
- ✓ OVSeg

Unsupervised Learning

- ✓ GroupViT
- ✓ MaskCLIP
- ✓ ReCo

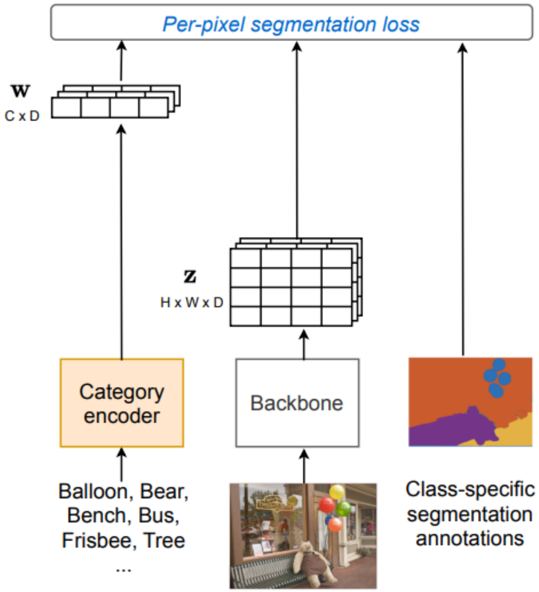
Related Works

Semi-supervised Learning

- ✔ Lseg
- ✔ OpenSeg
- ✔ OVSeg

segmentation mask + (image, text) pair

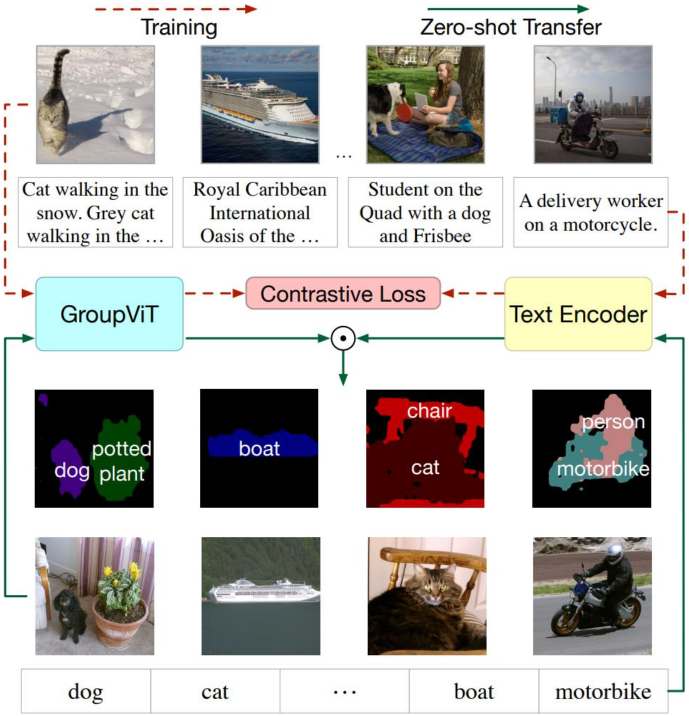
: dense annotation이 존재는 하나 각각이 어떤 class인지에 대한 정보는 알고 있지 않음



Golnaz Ghiasi et al., Scaling Open-Vocabulary Image Segmentation with Image-Level Labels, ECCV 2022

Related Works

only (image, text) pair

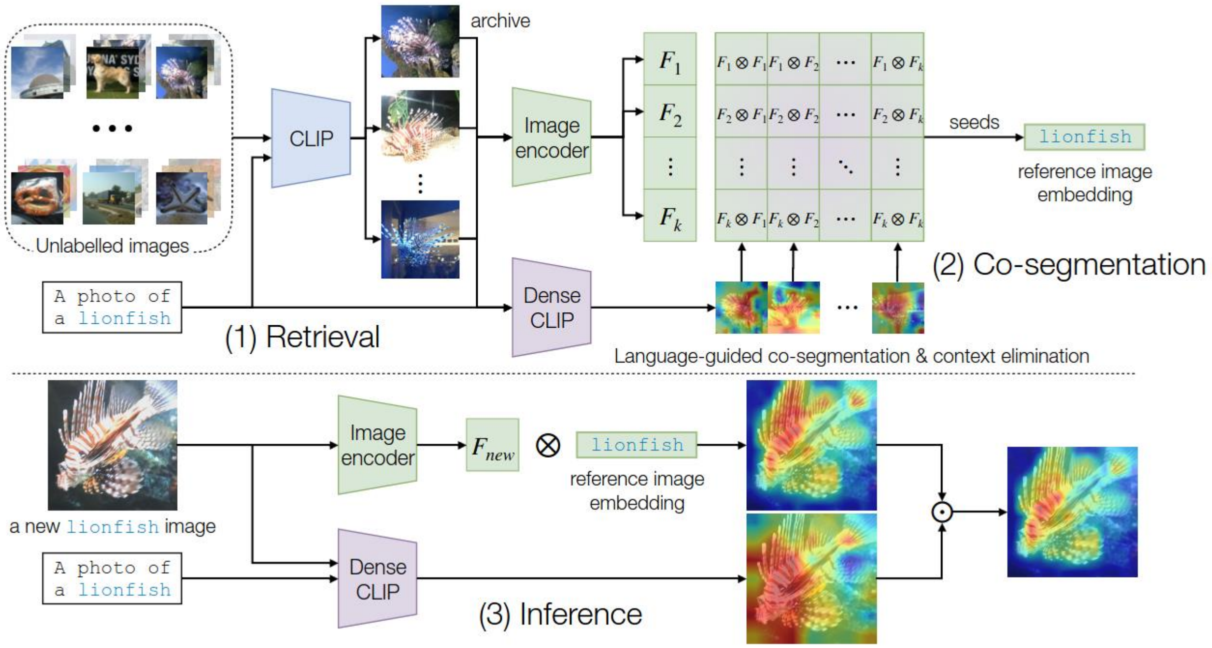


Unsupervised Learning

- ✔ GroupViT
- ✔ MaskCLIP
- ✔ ReCo

tributions

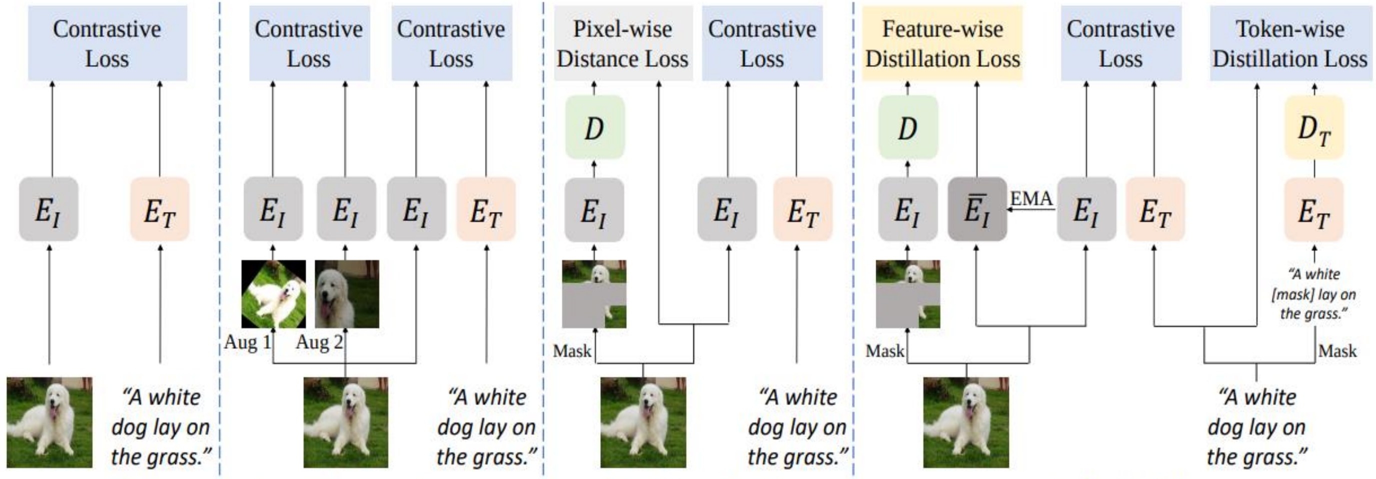
Previous Unsupervised Learning Methods



Dong, X., Bao, J., Zheng, Y., Zhang, T., Chen, D., Yang, H., Zeng, M., Zhang, W., Yuan, L., Chen, D., Wen, F., & Yu, N. (2022). MaskCLIP: Masked Self-Distillation Advances Contrastive Language-Image Pretraining. ArXiv. /abs/2208.12262

tributions

Previous Unsupervised Learning Methods



(a) CLIP (b) + Contrastive Learning (c) + Mask Image Modeling (d) MaskCLIP

Shin, G., Xie, W., & Albanie, S. (2022). ReCo: Retrieve and Co-segment for Zero-shot Transfer. ArXiv. /abs/2206.07045

tributions

Existing Methods: Indirectly address region-level alignment problems by learning image level alignment

TCL (Ours): propose a novel region-level objective

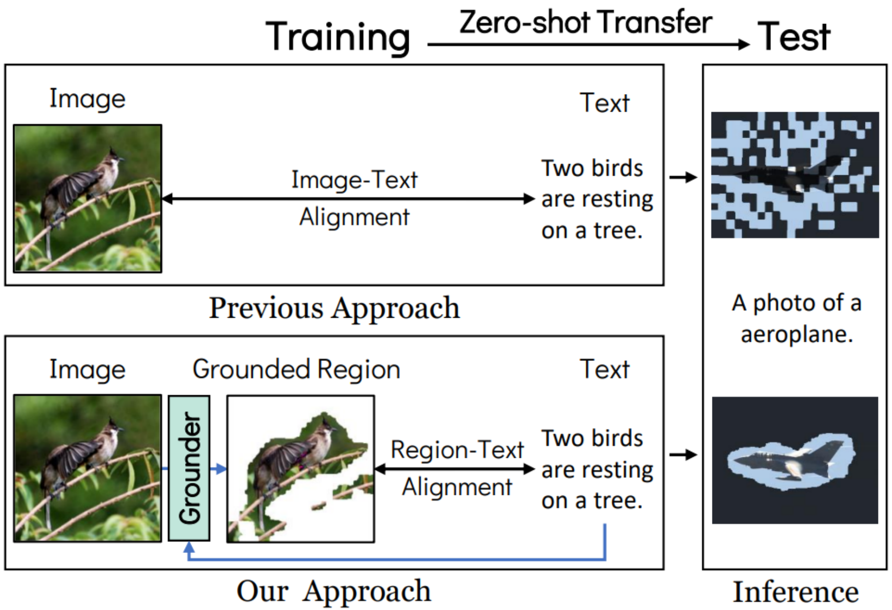


Figure 2. **A conceptual comparison between the previous approach and ours.** Open-world segmentation is typically achieved through region-text alignment, which involves matching region features and text embeddings. However, previous methods learn image-text alignment during training, thus suffering from the alignment-level discrepancy between training and testing. In contrast, our method facilitates end-to-end learning of region-text alignment with only image-text pairs.

Introduction | Related Works | Contributions | Methods | Experiments | Conclusions

Q&A

Methods: TCL

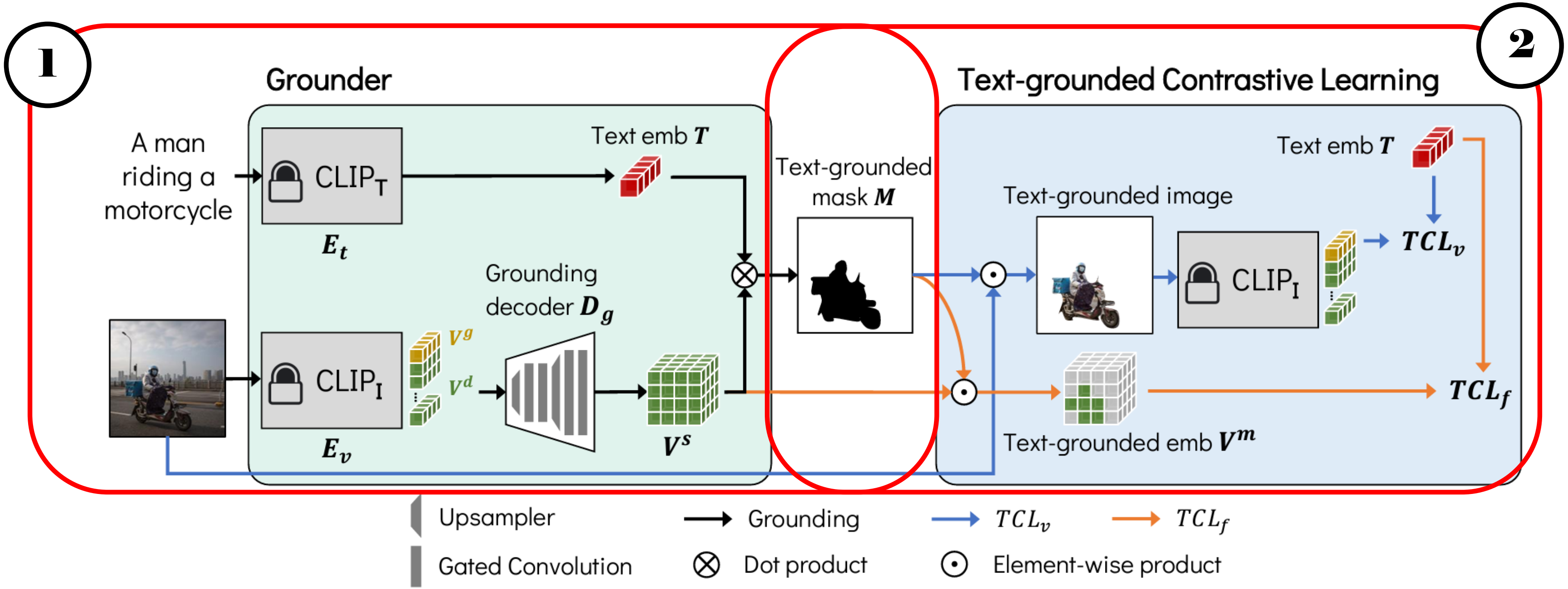
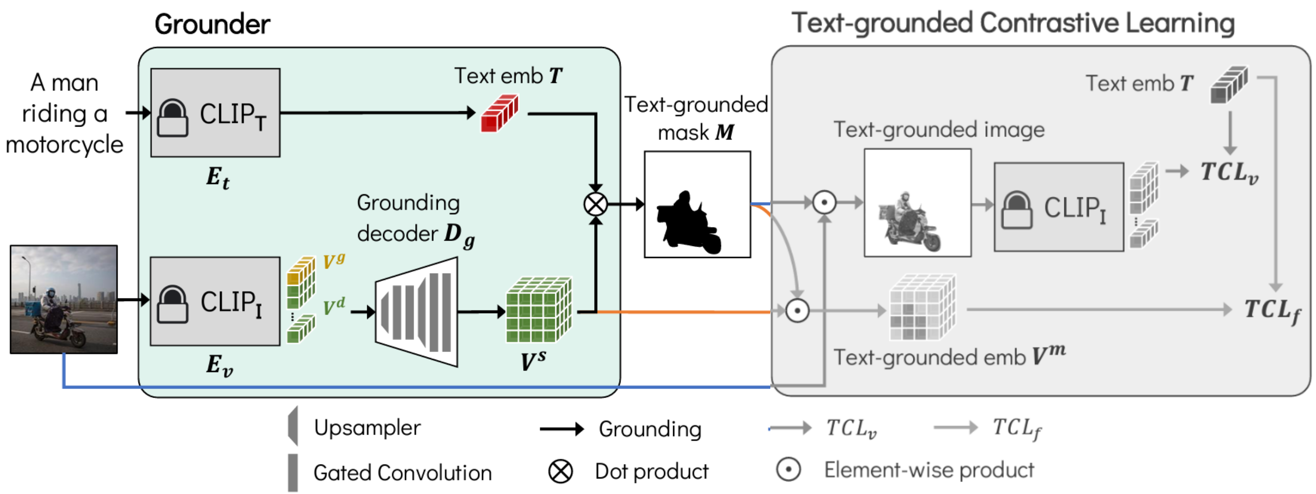


Figure 3. **Overall training pipeline of TCL.** The proposed TCL framework first obtains text-grounded masks using a grounder and then learns the grounder using text-grounded contrastive learning. By incorporating the text grounding process with contrastive learning, our framework can directly learn region-text alignment that is required for precise segmentation. $\mathrm{CLIP_T}$ and $\mathrm{CLIP_I}$ indicate CLIP text and image encoders, respectively. CLIP encoders are frozen and we train the grounding decoder only. After training, the TCL block is discarded, and only the Grounder block is used to generate the text-grounded segmentation mask for inference.

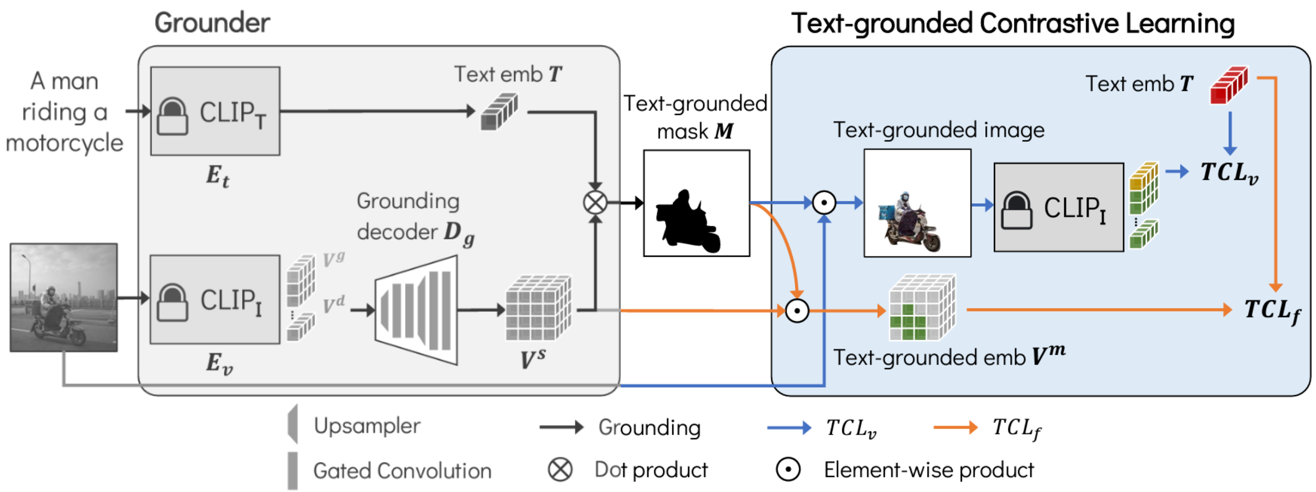
Methods: TCL



Grounder는 입력한 텍스트에 해당하는 영역, 즉 text-grounded mask(M)을 찾아내는 과정

- ✓ CLIP Text Encoder (CLIP$_T$): CLIP의 텍스트 인코더를 고정(froze)시키고 사용
- ✓ CLIP Image Encoder (CLIP$_I$): CLIP의 이미지 인코더를 고정(froze)시키고 사용
- ✓ Grounding Decoder (D$_g$): 이미지 인코더를 통해서 얻은 축소된 차원의 임베딩 벡터를 디코딩하는 역할. Grounding Decoder의 결과로 얻은 벡터와 텍스트 임베딩 벡터 간의 Dot product를 적용한 결과가 올바른 text-grounded mask를 생성하도록 학습된다.

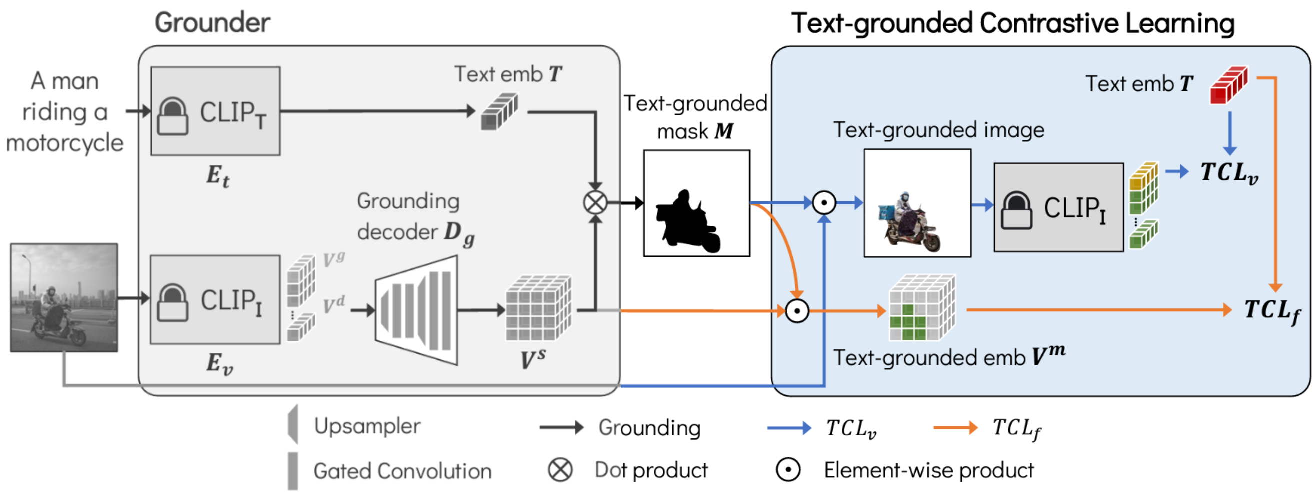
Methods: TCL



Grounder 단계를 거쳐서 얻은 text-grounded mask (***M***)를 활용하여 아래 두 가지를 수행하는 단계

✓ **TCLv:** image-level의 contrastive learning
✓ **TCLf:** feature-level의 contrastive learning

Methods: TCL

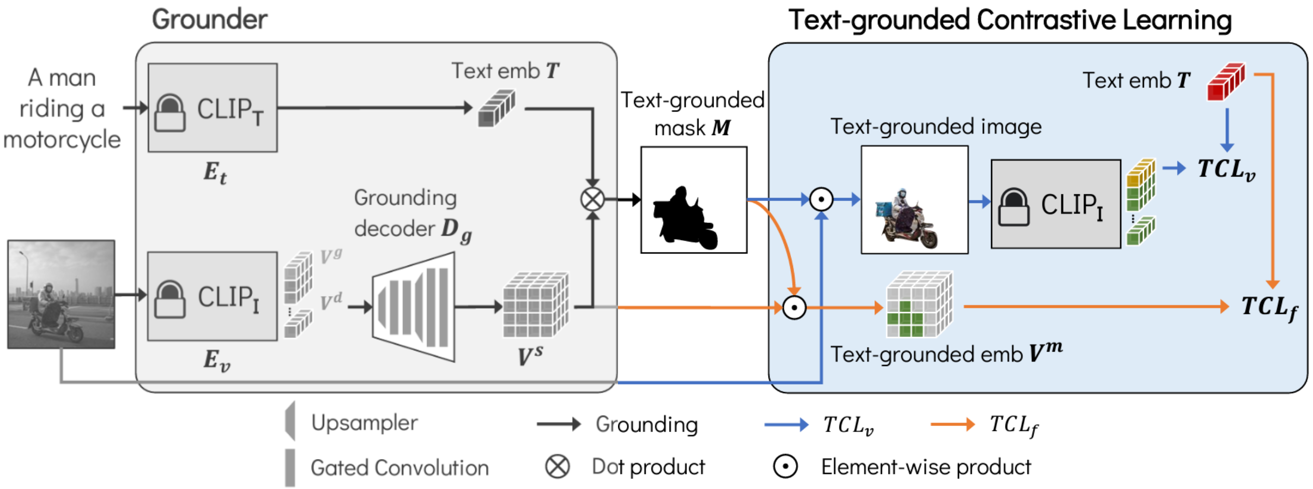


TCL

TCLv: image-level의 contrastive learning ⬅ Consider positive Text-grounded masks

1. Text-grounded mask와 원본 이미지를 elemental-wise product하여 Text-grounded Image를 얻어낸다.
2. Text-grounded Image를 (frozen) CLIP 인코더에 통과시키고 text-grounded image embedding을 얻는다.
3. InfoNCE로 text-grounded image embedding과 text embedding 간의 similarity matrix를 계산한 것이 **TCLv**이 된다.

Methods: TCL

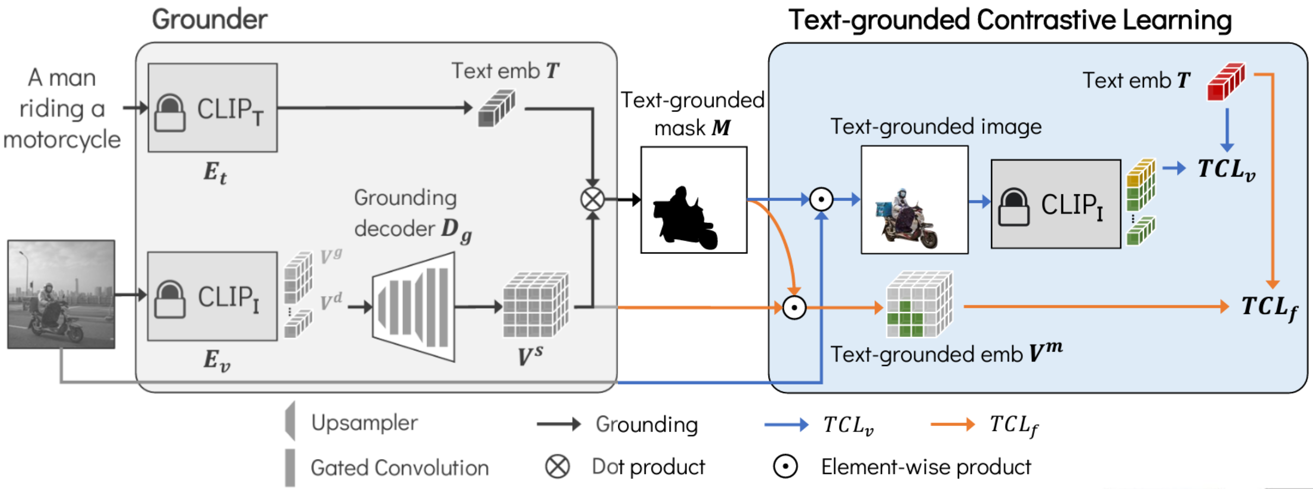


TCL

TCLf: feature-level의 contrastive learning ← **Giving negative feedbacks**

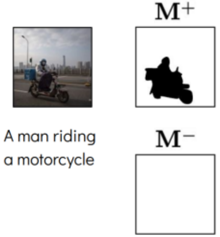
1. **Mi,j**에서 **i≠j**인 부분에 대해서 feature-level text-grounded image embedding을 계산하여 negative pair에 대한 임베딩을 얻는다.
2. 위 과정에서 얻은 text-grounded image embedding과 text embedding간에 InfoNCE를 계산해 similarity loss를 구한다.

Methods: TCL



Larea: 이미지 전체를 positive region으로 보는 것에 빠지는 것을 방지하기 위한 loss

Preventing Trivial Solution

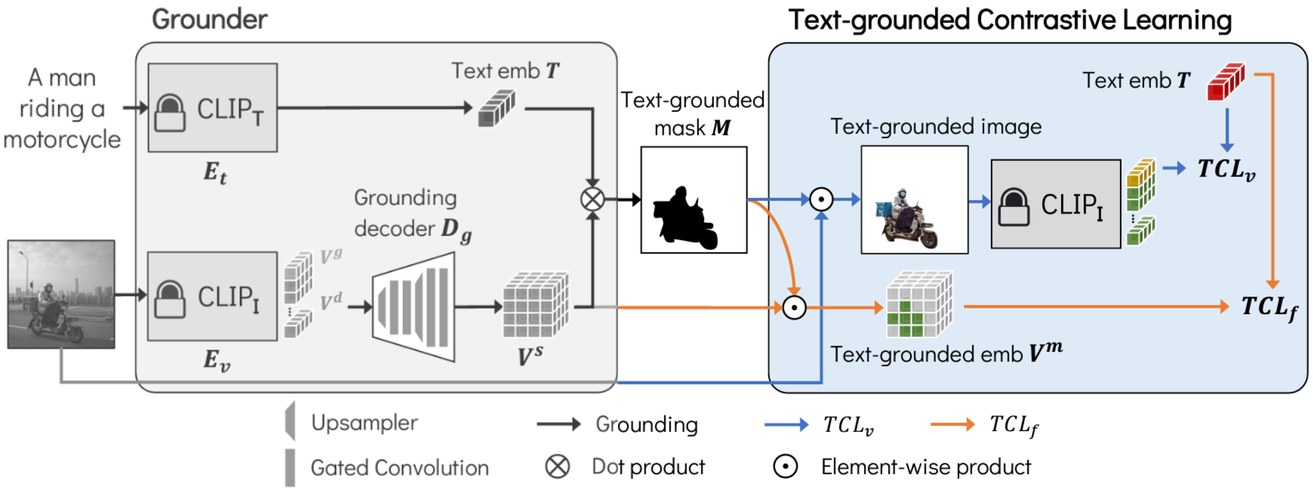


$$\mathcal{L}_{\text{area}} = \left\| p^+ - \mathbb{E}\left[\mathbf{M}^+\right] \right\|_1 + \left\| p^- - \mathbb{E}\left[\mathbf{M}^-\right] \right\|_1$$

Expectation of positive text-grounded mask area

Expectation of negative text-grounded mask area

Methods: TCL



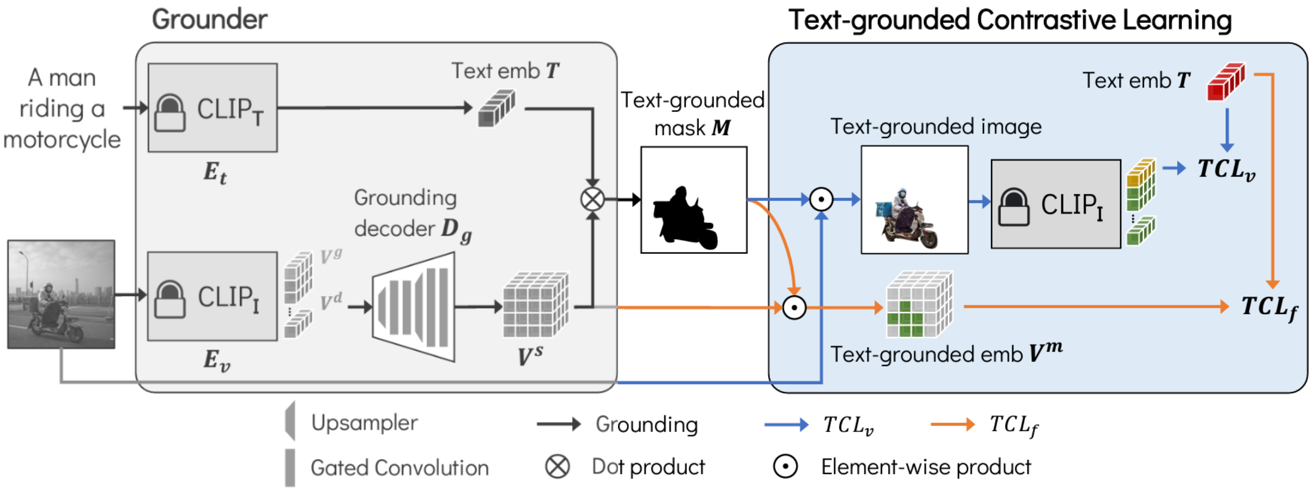
Loss$_{TV}$ (Total Variation Loss): 이미지에서 text에 해당하는 region들은 noisy하기보다는 smooth하다는 것을 발견하여 이와 같은 regularization을 추가

Smooth Regularization

$$\mathcal{L}_{\mathrm{tv}} = \|\mathbf{M}\|_{\mathrm{TV}} + \|\mathbf{V}^s\|_{\mathrm{TV}},$$

where $\|\cdot\|_{\mathrm{TV}}$ is the anisotropic TV norm.

Methods: TCL



$$\mathcal{L}_{\text{final}} = \overbrace{\lambda_{\text{TCL}}(\mathcal{L}_{\text{TCL}_v} + \mathcal{L}_{\text{TCL}_f}) + \lambda_{\text{area}}\mathcal{L}_{\text{area}}}^{\text{TCL losses}} + \overbrace{\lambda_{\text{tv}}\mathcal{L}_{\text{tv}}}^{\text{regularization}}$$

image-level TCL

feature-level TCL

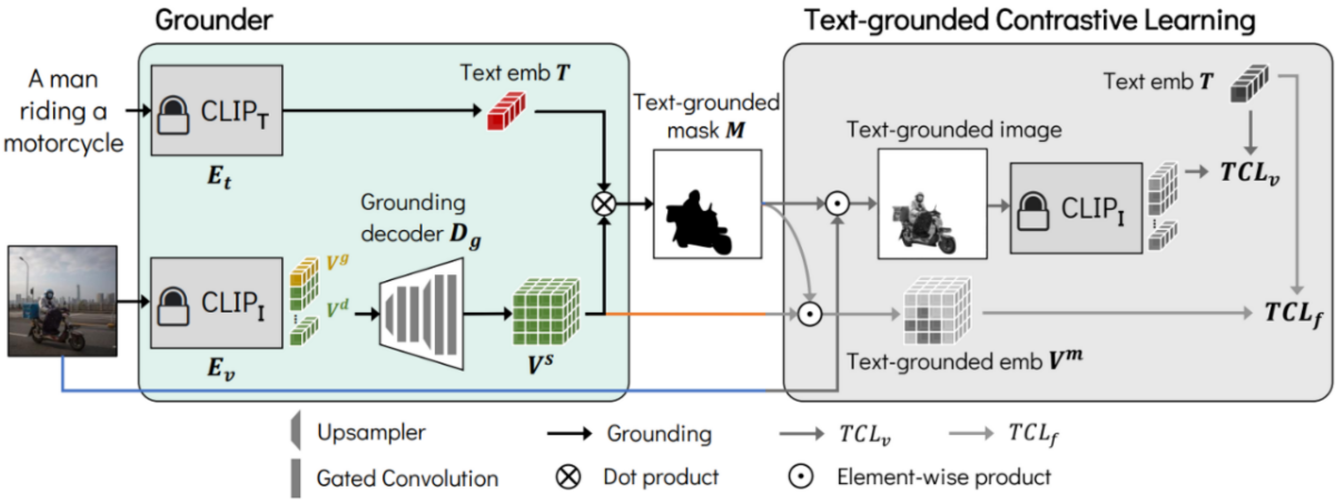
area TCL

smooth regularization

Methods: TCL

Inference Pipeline

- Training에서 사용되었던 TCL 블록은 사용하지 않고, Grounder의 결과를 inference 결과로 사용하게 된다.
- 즉, Text-grounded mask ($\boldsymbol{M}$)이 test 시의 마스크가 된다



Introduction Related Works Contributions Methods Experiments Conclusions

Q&A

Experiments

Implementation Details

- **Grounder:** CLIP ViT-B/16 (224 x 224 input image with 16 x 16 patch size)
- **Grounding Decoder:** four gated convolution blocks with two upsampling interpolations + pixel-adaptive mask refinement (PAMR) for mask refinement
- **Training Dataset:** CC3M + 12M datasets

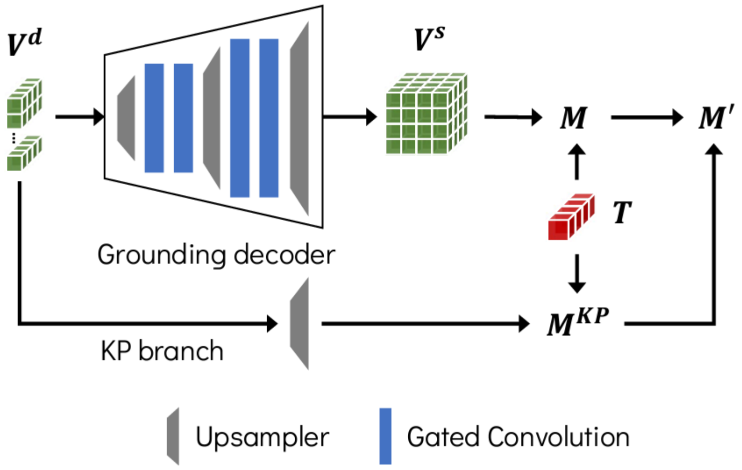


Figure 7. **Architecture of the grounding decoder.** The knowledge preservation (KP) branch serves to preserve pre-trained knowledge intact.

Experiments

Implementation Details

- **Grounder:** CLIP ViT-B/16 (224 x 224 input image with 16 x 16 patch size)
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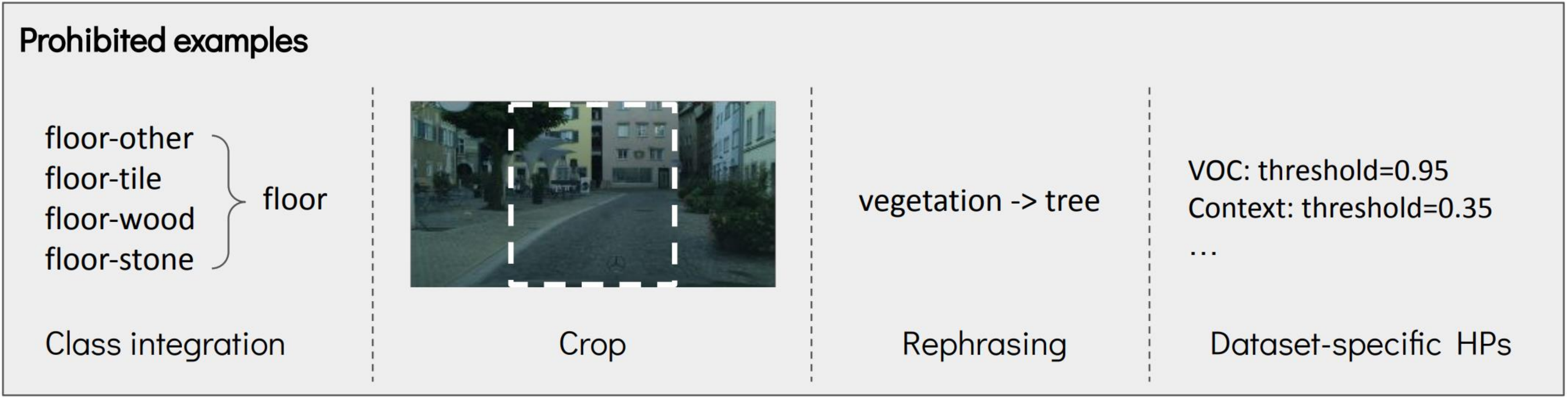


Experiments

Zero-shot Transfer to Semantic Segmentation

- •GroupViT: VOC, VOC20, 그리고 COCO-object와 같이 object-oriented dataset에서는 높은 성능을 보임. 하지만 stuff classes(잔디, 하늘과 같이 배경인 classes)에 대해서는 낮은 성능을 보임
- •MaskCLIP: Context, Context59, COCO-stuff와 같이 stuff-oriented dataset에서 높은 성능을 보임. MaskCLIP의 refinement technique들은 대부분 좋은 성능을 보였으나 Cityscape에서는 낮아지는 결과를 보였는데 이는 heuristic refinement 방법들의 한계를 보여주는 것이라고 설명
- •ReCo: MaskCLIP 방법론들에 비해서 VOC와 VOC20과 같이 background를 포함한 데이터셋에서 많은 성능 하락이 있었음.
- •TCL (Ours): 가장 높은 성능을 보임. train 데이터셋과 test 데이터셋 간의 간극을 최소화하며 잘 학습된 것이라고 말할 수 있음. 또한 background도 잘 구별해 냈는데, 이는 data-driven manner로 학습되었기 때문에 heuristic post-processing보다 좋은 성능을 낸 것으로 설명

Methods	with background class VOC	Context	Object	without background class VOC20	Context59	Stuff	City	ADE	Avg.
GroupViT (YFCC)	49.5	19.0	24.3	74.1	20.8	12.6	6.9	8.7	27.0
GroupViT (RedCaps)	50.4	18.7	27.5	79.7	23.4	15.3	11.1	9.2	29.4
MaskCLIP†	29.3	21.1	15.5	53.7	23.3	14.7	21.6	10.8	23.7
MaskCLIP	38.8	23.6	20.6	74.9	26.4	16.4	12.6	9.8	27.9
ReCo	25.1	19.9	15.7	57.7	22.3	14.8	21.1	11.2	23.5
TCL (Ours)	**55.0** (+4.6)	**30.4** (+6.8)	**31.6** (+4.1)	**83.2** (+3.5)	**33.9** (+7.5)	**22.4** (+6.0)	**24.0** (+2.4)	**17.1** (+5.9)	**37.2** (+7.8)

Table 1. **Zero-shot segmentation performance comparison on 8 semantic segmentation datasets.** mIoU metric is used in every experiment. We highlight **the best** and second-best results. MaskCLIP† indicates their baseline method without additional refinement techniques. The YFCC and RedCaps of GroupViT indicate their training datasets in addition to CC12M. Each dataset abbreviation stands for VOC: PASCAL VOC, Context: PASCAL Context, Object: COCO-Object, Stuff: COCO-Stuff, City: Cityscapes, ADE: ADE20K.

Experiments

Qualitative Results

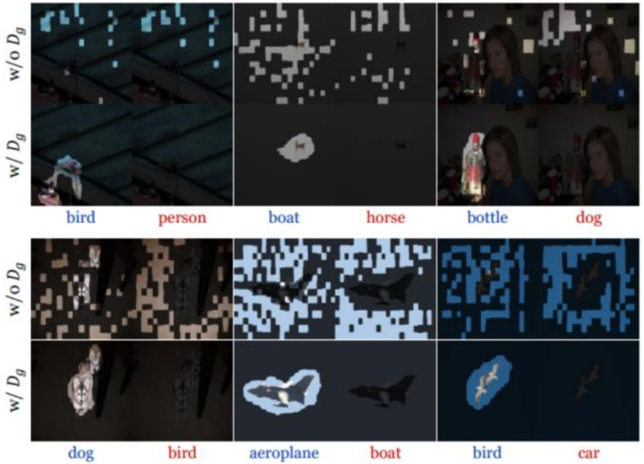
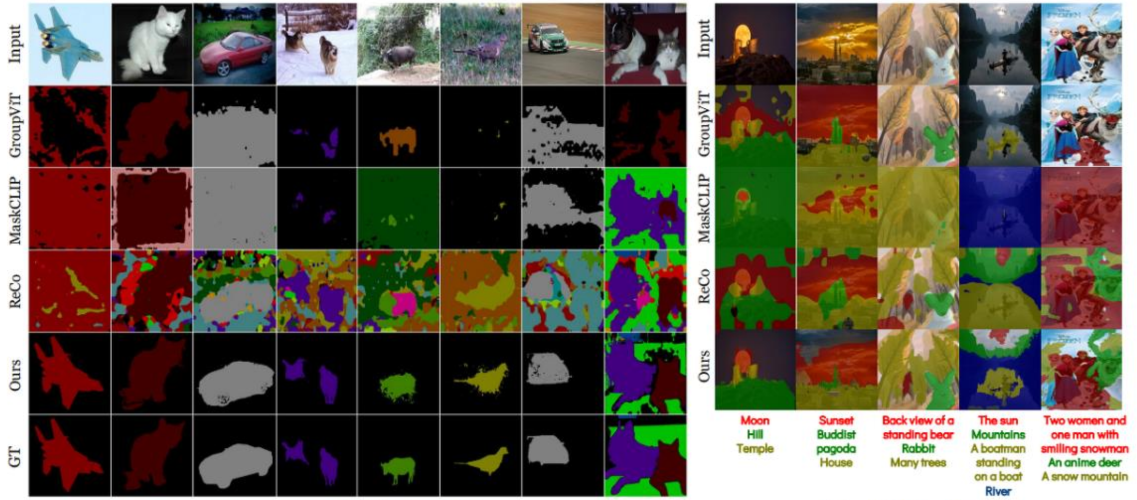


Figure 4. **Visualization of the generated text-grounded masks.** "w/o D_g" rows show the generated text-grounded masks without the grounding decoder (D_g), *i.e.*, CLIP dense features $\mathbf{V}^d$ are used instead of pixel-level dense embeddings $\mathbf{V}^s$. "w/ D_g" rows show the results with the grounding decoder. Each image is compared using both positive (blue) and negative (red) prompts. The results show that the grounder accurately and finely captures the text-described region with less noise via the grounding decoder.



(a) **Examples in PASCAL VOC.** (b) **Examples in the wild.**

Figure 5. (a) The comparison shows the error types of each method in the VOC dataset. GroupViT tends to make an error on a large group rather than noisy results. ReCo suffers from segmentation of the background region. MaskCLIP tends to fail at capturing the target area precisely. (b) shows results on the wild web images and free-form texts. Texts used as target classes are shown at the bottom of the images.

Experiments

Ablation Study

- **Baseline:** initial model before training based on MaskCLIP
- **+ Decoder:** add grounding decoder
- **+ TCL:** 모든 TCL loss를 다 더한 것

	Method	VOC20
A	Baseline	53.2
B	+ Decoder	52.3
C	+ TCL	**77.4**

(a) **Baseline to TCL.**

	TCL_v	TCL_f	$\mathcal{L}_{\text{area}}$	CL	VOC20
D				✔	61.1
E	✔		✔		74.6
F		✔	✔		76.0
C	✔	✔	✔		**77.4**
G	✔	✔	✔	✔	75.6
H	✔	✔			67.1

(b) **TCL losses.**

λ_{TCL}	VOC20
0.0	-
0.01	76.8
0.1	**77.4**
1.0	68.2

(c) $\mathcal{L}_{\text{TCL}}$

λ_{area}	VOC20
0.0	67.1
0.04	69.5
0.4	**77.4**
4.0	76.7

(d) $\mathcal{L}_{\text{area}}$

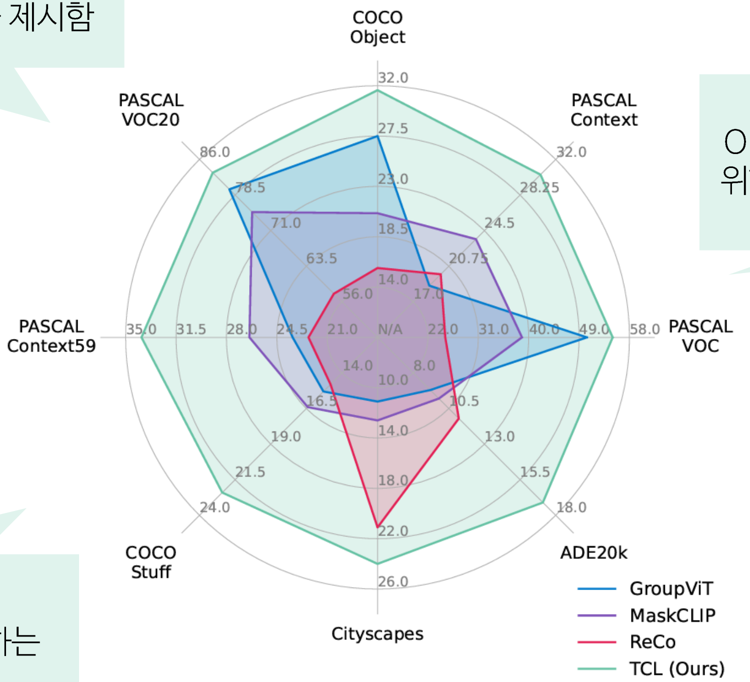
λ_{tv}	VOC20
0.0	73.8
0.1	75.2
1.0	**77.4**
10.0	70.7

(e) $\mathcal{L}_{\text{tv}}$

Table 2. **Ablation studies on TCL losses and hyperparameters.** Refinement techniques are not applied to reveal the effect of each loss function clearly. Default settings are marked in gray .

Conclusions

Region-text alignment를 통해서 train-test discrepancy 없이 unsupervised open-world segmentation 새로운 방법론을 제시함



Open-world segmentation을 평가하기 위한 unified evaluation protocol을 제시

8개의 데이터셋에서 zero-shot segmentation을 수행했을 때 현존하는 모델 중 가장 좋은 성능을 보임

Q&A